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On Queues in Heavy Traffic

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Summary

We say that a single server queue is in "heavy traffic" when the traffic intensity ρ is less than, but very near, unity. Then the equilibrium waiting time w will be large, and it is proved that, under certain conditions, the distribution of $(1-\rho)w$ tends, as ρ tends to 1, to a negative exponential distribution. The conditions under which this result holds are fairly weak, and, in particular, it is not necessary to make the assumption of independent inter-arrival times.

1. Introduction

One of the main conclusions to be drawn from the large amount of research that has been carried out in recent years on the theory of queues is that, in order to obtain explicit results, it is necessary to make very strong assumptions about the statistical nature of the processes involved; assumptions which may be satisfied only approximately, if at all, in practice.

It is therefore of some interest to ask if there are any results which hold under somewhat weaker conditions. It is not to be expected that these will be results valid in general, but rather that they will hold in some limiting case. The limiting case to be considered here is that usually known as "heavy traffic".

A common way of formulating queueing problems with a single server is in terms of a traffic intensity ρ defined as the average number of customers arriving during a service period. For $\rho > 1$ the queue will be stable, while for $\rho \geqslant 1$ its length will increase indefinitely as time goes on. In particular, when $\rho < 1$ an equilibrium waiting-time distribution will exist, and as ρ approaches 1 from below, this waiting time will become long. This situation, when ρ is less than, but near, unity, we describe as "heavy traffic". The main conclusion of the present paper may roughly be stated as follows: *If w is the equilibrium waiting time, then the distribution of $(1-\rho)w$ is asymptotically negative exponential as ρ tends to* 1.

The parameter of this exponential distribution is given in terms of the means and variances of the input and service processes. In the particular case of the queue GI/G/1 the result has been proved by the author (1961).

We shall first indicate heuristically why this result is to be expected, and what conditions it is necessary to impose. A rigorous proof is then given for a wide class of queues, and this is shown to include a number of cases of interest. The corresponding analysis for the queue length and the busy period is also briefly discussed. All the work in this paper may be re-interpreted in the usual way in terms of random walks or semi-infinite dams.

2. Formulation and Preliminary Results

The system to be considered is a single server queue at which customers $C_0, C_1, \ldots$ arrive and are served in the order of their arrival. C_0 arrives at $t = 0$ and finds the

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server free. The arrival time, service time and waiting time (excluding service) of the customer C_n are respectively T_n, s_n and w_n. We write

$$t_n = T_{n+1} - T_n, \quad u_n = s_n - t_n. \tag{1}$$

Clearly $w_0 = 0$. Lindley (1952) has shown that the sequence $\{w_n\}$ is determined by

$$w_{n+1} = (w_n + u_n)^+, \tag{2}$$

where $x^+ = \max(x, 0)$.

We make the following assumption about the statistical nature of the process $\{u_n\}$: $\mathscr{A}$: $\{u_n\}$ forms a (strictly) stationary, metrically transitive process, with finite mean $-\mu < 0$.

Under this assumption, Loynes (1962a) has shown that the distribution function of w_n tends to a limit (the equilibrium waiting-time distribution) which is the distribution function of the random variable w defined almost surely by

$$w = \sup_{n \geqslant 0} \sum_{r=1}^{n} u_{-r}. \tag{3}$$

(We make the convention that $\sum_{r=1}^{0} = 0$, so that $w \geqslant 0$.)

One of the main objectives of queueing theory has been the evaluation of the distribution of w. In the case when the u_n are independent, this was done by Smith (1953), using the method of Hopf and Wiener, and extending earlier results of Erlang, Pollaczek, Kendall and others.

We shall here be interested in the limiting case when μ is small. More precisely, suppose that, for each $\alpha > 0$, we define a queue Q_α for which the sequence $u_n(\alpha)$ satisfies $\mathscr{A}$, for which the variance $\sigma^2(\alpha)$ of $u_n(\alpha)$ exists, and for which

$$\alpha = \mu(\alpha)/\sigma(\alpha). \tag{4}$$

Then, if

$$z(\alpha) = \mu(\alpha)\, w(\alpha)/\sigma^2(\alpha), \tag{5}$$

we shall prove that, under certain conditions,

$$P\{z(\alpha) \geqslant z\} \to e^{-2z} \quad (\alpha \to 0, z \geqslant 0). \tag{6}$$

Thus, for $\alpha = \mu/\sigma$ small, the waiting time is approximately exponentially distributed with mean

$$E(w) \simeq \operatorname{var}(u)/-2E(u). \tag{7}$$

Before exhibiting a class of queues for which the result can be proved, we give in the next section a heuristic derivation to indicate why it may be expected to hold under fairly weak conditions.

The approximate expression (7) for the mean waiting time is in fact an upper bound when the u_n are independent. In fact, if this is so, we have

$$\begin{aligned} 0 &\leqslant \operatorname{var}(w_n + u_n)^- \\ &= E\{(w_n + u_n)^-\}^2 - \{E(w_n + u_n)^-\}^2 \\ &= E(w_n + u_n)^2 - E\{(w_n + u_n)^+\}^2 - \{E(w_n + u_n) - E(w_n + u_n)^+\}^2 \\ &= E\{w_n^2\} + 2E\{w_n\}.E\{u\} + E\{u^2\} - E\{w_{n+1}^2\} - \{E(w_n + u_n - w_{n+1})\}^2. \end{aligned}$$

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Letting n tend to infinity,

$$0 \leqslant E\{w^2\}+2E\{w\}.E\{u\}+E\{u^2\}-E\{w^2\}-E\{u\}^2$$
$$= 2E\{w\}.E\{u\}+\operatorname{var}(u).$$

Since $Eu<0$ we obtain

$$E(w) \leqslant \operatorname{var}(u)/\{-2E(u)\}. \tag{7'}$$

3. Heuristic Treatment

Using $[\phi]$ to denote the integral part of a number ϕ, consider the process

$$x_\alpha(t) = \frac{\alpha}{\sigma(\alpha)} \sum_{n=1}^{[\alpha^{-2}t]} u_{-n}(\alpha). \tag{8}$$

Then, from (3), the equilibrium waiting time $w(\alpha)$ in Q_α is given by

$$w(\alpha) = \alpha^{-1}\,\sigma(\alpha) \sup_{t\geqslant 0} x_\alpha(t),$$

so that

$$z(\alpha) = \mu(\alpha)\,w(\alpha)/\sigma^2(\alpha) = \sup_{t\geqslant 0} x_\alpha(t). \tag{9}$$

Since we wish to consider the limiting distribution of $z(\alpha)$ as $\alpha\to 0$, it is natural to consider the limiting joint distributions of the process $x_\alpha(t)$. Suppose first that the u_n are independent. Then

$$x_\alpha(t_1)-x_\alpha(t_2) \quad (t_1>t_2)$$

is a sum of independent random variables, and it is easy to see that it has been normalized to have mean

$$-(t_1-t_2)+o(1)$$

and variance

$$(t_1-t_2)+o(1),$$

as $\alpha\to 0$. By the central limit theorem, it is asymptotically normally distributed. This type of argument shows that the joint distributions of $x_\alpha(t)$ tend to those of a Brownian process $x(t)$ with mean $-t$ and variance t.

That this result could be expected to hold more generally follows from the fact that the central limit theorem can be extended to dependent random variables. Bernstein (1927) and Loève (1960, chap. 8) have conditions for this extension to be valid; these take the form of inequalities on certain conditional moments of the sequence $\{u_n\}$. Unfortunately, however, these do not seem to be readily applicable to the problems that naturally arise in queueing theory, and it is more convenient to proceed directly.

If the joint distributions of $x_\alpha(t)$ do, in fact, tend to those of $x(t)$, we should expect that the distribution of the supremum $z(\alpha)$ might tend to that of

$$z_0 = \sup_{t\geqslant 0} x(t). \tag{10}$$

(To prove this for a particular process is equivalent to establishing the validity of the interchange of two limiting operations which do not in general commute.)

We may now invoke a result of Baxter and Donsker (1957) that z_0 has a negative exponential distribution with mean $\frac{1}{2}$, and we thus obtain equation (6).

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This argument is not easy to make rigorous, and we shall therefore proceed in the next section to establish the result by a different method.

Before doing this, we note that the time scale has been changed by defining $x_\alpha(t)$ as a sum over $[\alpha^{-2}t]$ members of the sequence $\{u_n(\alpha)\}$. This means that the time taken to achieve equilibrium is of the order of α^{-2}, and therefore, when α is small, the system settles down only slowly into a steady state. The results of the present analysis only refer to the waiting time when that steady state has, in fact, been reached.

Another consequence of the heuristic theory is that the quantity that must be small for a heavy traffic situation to exist is not

$$1-\rho = -E(u)/E(t)$$

but

$$\alpha = -E(u)/\mathrm{var}(u).$$

For example, suppose that we modify a queue by adding a fixed quantity B to both s_n and t_n (for all n).

The new value of the traffic intensity is

$$\rho_B = \{E(s)+B\}/\{E(t)+B\},$$

and this tends to 1 as $B\to\infty$. The modification does not, however, lead to a heavy traffic situation, as u_n, and hence w, is unchanged. In fact, from the point of view of heavy traffic, α is a much more natural dimensionless parameter than ρ.

If, for a sequence of queues Q_α, $z(\alpha)$ is asymptotically exponentially distributed with mean $\frac{1}{2}$, we shall say that Q_α has the *heavy traffic property*.

4. A Class of Queues with the Heavy Traffic Property

In an earlier paper (Kingman, 1961) it was shown that, even when the $u_n(\alpha)$ are independent, the argument is complicated by the fact that the distributions may behave very irregularly as $\alpha\to 0$. Since these irregularities owe nothing to the structure of the queue, we shall here simplify the situation by assuming that the normalized process

$$\{u_n(\alpha)+\mu(\alpha)\}/\sigma(\alpha)$$

does not depend on α. It is not difficult to modify the discussion to more general situations.

If this simplification is made, we have

$$u_{-n}(\alpha) = \sigma(\alpha)(y_n-\alpha), \tag{11}$$

where $\{y_n\}$ is a stationary, metrically transitive sequence with zero mean and unit variance. Equations (3) and (5) lead to

$$z(\alpha) = \alpha \sup_{n\geqslant 0} \sum_{r=1}^{n} (y_r-\alpha). \tag{12}$$

We shall say that the sequence $\{y_r\}$ satisfies condition $\mathscr{C}$ if the distribution of

$$\alpha \sup_{n\geqslant 0} \sum_{r=1}^{n} (y_r-\alpha)$$

tends to a negative exponential distribution with mean $\frac{1}{2}$. Equation (12) leads immediately to

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Theorem 1. A sequence of queues Q_α with $u_n(\alpha)$ defined as in (11) has the heavy traffic property if and only if $\{y_n\}$ satisfies $\mathscr{C}$.

The problem is therefore reduced to that of determining which stationary sequences satisfy $\mathscr{C}$. The case in which the sequence is independent has already been dealt with, though in a rather different way (Kingman, 1961), and the proof will therefore merely be sketched. The assumption of an analytic service time characteristic function is probably unnecessarily strong.

Theorem 2. ($\mathscr{C}$ for independent sequences.) Let $\{y_n\}$ be a sequence of identically distributed, independent random variables with zero mean and unit variance, and suppose that, for some positive b,

$$E(e^{by}) < \infty. \tag{13}$$

Then
$$P\left\{\alpha \sup_{n \geqslant 0} \sum_{r=1}^{n} (y_r - \alpha) \leqslant z\right\} = P\{y_n \leqslant \alpha n + \alpha^{-1} z, \text{ all } n\} \to 1 - e^{-2z} \tag{14}$$

as
$$\alpha \to 0 \ (z \geqslant 0).$$

Sketch of proof: We first proceed as in Spitzer (1957, Theorem 3) to show that

$$(1-x) \sum_{n=0}^{\infty} E\left[\exp\left\{-\lambda \sup_{r \leqslant n} \alpha \sum_{j=1}^{r} (y_j - \alpha)\right\}\right] x^n$$
$$= \exp\left[\frac{1}{2\pi} \int_0^x \frac{d\tau}{1-\tau} \int_{-\infty}^{\infty} \frac{\phi(\alpha\xi) e^{-i\alpha^2\xi} - 1}{1 - \tau\phi(\alpha\xi) e^{-i\alpha^2\xi}} \frac{\lambda d\xi}{\xi(\xi - i\lambda)}\right]$$

where $\phi(\xi) = E(e^{i\xi y})$. Hence

$$\log E[\exp\{-\lambda z(\alpha)\}] = \frac{1}{2\pi} \lim_{x \to 1-} \int_0^x \frac{d\tau}{1-\tau} \int_{-\infty}^{\infty} \frac{\phi(\alpha\xi) e^{-i\alpha^2\xi} - 1}{1 - \tau\phi(\alpha\xi) e^{-i\alpha^2\xi}} \frac{\lambda d\xi}{\xi(\xi - i\lambda)}.$$

We now change the contour of integration to $(-ic-\infty, -ic+\infty)$, which is valid for suitable c by (13). This enables us to interchange orders of integration, giving, after some manipulation,

$$\log E[\exp\{-\lambda z(\alpha)\}] = -\frac{\lambda}{2\pi} \int_{-ic-\infty}^{-ic+\infty} \log\left\{\frac{\phi(\alpha\xi) e^{-i\alpha^2\xi} - 1}{-i\alpha^2 \xi}\right\} \frac{d\xi}{\xi(\xi - i\lambda)}.$$

It is not difficult to see that, as $\alpha \to 0$, the expression in the curly brackets tends to $1 - \frac{1}{2} i\xi$, so that (using the dominated convergence theorem)

$$\log E[\exp\{-\lambda z(\alpha)\}] \to -\frac{\lambda}{2\pi} \int_{-ic-\infty}^{-ic+\infty} \log(1 - \tfrac{1}{2} i\xi) \frac{d\xi}{\xi(\xi - i\lambda)},$$

which may readily be evaluated by contour integration as

$$-\log(1 + \tfrac{1}{2}\lambda).$$

Hence $z(\alpha)$ has an asymptotically exponential distribution with mean $\frac{1}{2}$ as α tends to 0.

Roughly speaking, the meaning of this theorem is that, in a random walk with small negative drift, the maximum value attained is inversely proportional to the drift.

In the next theorem, the class of processes known to satisfy $\mathscr{C}$ is very considerably enlarged by showing that we can modify a process in quite a general way without affecting $\mathscr{C}$.

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Theorem 3. Let $\{x_n\}$ be any process, and let $\{v_n\}$ be a process such that

$$P(|v_n| \geqslant A) \leqslant P(V \geqslant A) \quad (\text{all } A), \tag{15}$$

where V is a random variable with finite second moment.

Define a new sequence $\{y_n\}$ by

$$y_n = x_n + v_n - v_{n-1}, \tag{16}$$

and write

$$\xi(\alpha) = \alpha \sup_{n \geqslant 0} \sum_{r=1}^{n} (x_r - \alpha), \quad \eta(\alpha) = \alpha \sup_{n \geqslant 0} \sum_{r=1}^{n} (y_r - \alpha),$$

assuming these to be finite with probability one. Then if $\xi(\alpha)$ has a limiting distribution as $\alpha \to 0$, so has $\eta(\alpha)$, and these two distributions are the same. In particular, $\{y_n\}$ satisfies $\mathscr{C}$ if and only if $\{x_n\}$ does.

Lemma. Let ϕ be a random variable with $E(\phi)^2$ finite. Then, as c tends to infinity,

$$E(\phi - c)^+ = o(c^{-1}). \tag{17}$$

Proof. Since $E(\phi - c)^+$ is a decreasing function of c it is sufficient to prove the result for integral c. Let N be the integer next above ϕ, so that $E(N^2) < \infty$, and put

$$a_c = E(\phi - c)^+.$$

Then

$$\begin{aligned}
\sum_{c=0}^{n} a_c &= E\left\{\sum_{c=0}^{n} (\phi - c)^+\right\} \\
&\leqslant E\left\{\sum_{c=0}^{n} (N - c)^+\right\} \\
&= E\left\{\sum_{c=0}^{\min(N,n)} (N - c)\right\} \\
&\leqslant E\left\{\sum_{c=0}^{N} (N - c)\right\} \\
&= E\{\tfrac{1}{2}N(N+1)\} < \infty.
\end{aligned}$$

Hence Σa_c converges. But $\{a_c\}$ is a decreasing sequence, and hence $a_c = o(c^{-1})$ as c tends to infinity through integer values.

Proof of Theorem 3:

Let $\beta = \beta(\alpha)$ be a function of α such that $\beta < \alpha$ and $\beta = o(\alpha)$ as $\alpha \to 0$. Then

$$\begin{aligned}
\eta(\alpha) &= \alpha \sup \sum_{1}^{n} (x_r + v_r - v_{r-1} - \alpha) \\
&= \alpha \sup \left\{\sum_{1}^{n} [x_r - (\alpha - \beta)] + [v_n - v_0 - \beta n]\right\} \\
&\leqslant \alpha \sup \sum_{1}^{n} [x_r - (\alpha - \beta)] + \alpha \sup (v_n - v_0 - \beta n) \\
&= \frac{\alpha}{\alpha - \beta} \xi(\alpha - \beta) + \alpha \sup (v_n - v_0 - \beta n).
\end{aligned}$$

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Similarly,

$$\xi(\alpha) \leqslant \frac{\alpha}{\alpha-\beta}\eta(\alpha-\beta)+\alpha\sup(-v_n+v_0-\beta n).$$

Hence it suffices to prove that

$$\alpha\sup(v_n-v_0-\beta n) \quad \text{and} \quad \alpha\sup(-v_n+v_0-\beta n)$$

tend to zero in probability as $\alpha\to 0$, for some choice of $\beta(\alpha) = o(\alpha)$. For any positive A, let

$$\begin{aligned} p(A,\alpha,\beta) &= P\{\alpha\sup(v_n-v_0-\beta n)\geqslant A\} \\ &= P(v_n-v_0\geqslant A\alpha^{-1}+\beta n, \text{ some } n) \\ &\leqslant \sum_{n=1}^{\infty} P(v_n-v_0\geqslant A\alpha^{-1}+\beta n). \end{aligned}$$

Now

$$\begin{aligned} P(v_n-v_0\geqslant B) &\leqslant P(v_n\geqslant \tfrac{1}{2}B)+P(v_0\leqslant -\tfrac{1}{2}B) \\ &\leqslant 2P(V\geqslant \tfrac{1}{2}B). \end{aligned}$$

Hence

$$\begin{aligned} p(A,\alpha,\beta) &\leqslant 2\sum_{n=1}^{\infty} P(V\geqslant \tfrac{1}{2}A\alpha^{-1}+\tfrac{1}{2}\beta n) \\ &= 2E(\text{number of } n \text{ with } V\geqslant \tfrac{1}{2}A\alpha^{-1}+\tfrac{1}{2}\beta n) \\ &\leqslant 2E\{\beta^{-1}(2V-A\alpha^{-1})^+\} \\ &= 4E(V-\tfrac{1}{2}A/\alpha)^+/\beta \\ &= 4\beta^{-1}G(A/\alpha), \qquad \text{say.} \end{aligned}$$

Exactly similarly, we have

$$P[\alpha\sup(-v_n+v_0-\beta n)\geqslant A]\leqslant 4\beta^{-1}G(A/\alpha).$$

It therefore suffices to prove that we can choose $\beta(\alpha)$ so that $\beta(\alpha) = o(\alpha)$, and such that, for each $A\geqslant 0$,

$$G(A/\alpha) = o\{\beta(\alpha)\},$$

as α tends to zero.

Take

$$\beta(\alpha) = \left[\alpha\sum_{n=1}^{\infty} 2^{-n}G\{1/(n\alpha)\}\right]^{\frac{1}{2}},$$

which is finite since G is bounded by EV. Since, by the lemma, $G(x) = o(1/x)$ as $x\to\infty$, Tannery's theorem shows that

$$\sum_{n=1}^{\infty} 2^{-n}nG\{1/(n\alpha)\}/(n\alpha)\to 0$$

as $\alpha\to 0$, and so $\beta(\alpha) = o(\alpha)$.

Also,

$$\beta(\alpha)\geqslant[\alpha 2^{-n}G\{1/(n\alpha)\}]^{\frac{1}{2}}$$

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so that, if n is the integer next above $1/A$,

$$G(A/\alpha) \leqslant G\{1/(n\alpha)\} \leqslant 2^n \beta^2(\alpha)/\alpha = o(\beta(\alpha)).$$

Hence the theorem is proved.

Corollary. If $\{x_n\}$ is an independent sequence satisfying the conditions of Theorem 2, then $\{y_n\}$ satisfies $\mathscr{C}$.

We shall now specialize the results in order to make them more immediately relevant to the structure of the queue. For this we require the concept of a "weakly dependent input".

Definition. A queue is said to have a *weakly dependent input* if the inter-arrival time t_n is given by

$$t_n = r_n + v_n - v_{n-1}, \tag{18}$$

where the r_n form an independent sequence of identically distributed variables with finite mean, and $\{v_n\}$ is a process satisfying, for all A,

$$P(|v_n| \geqslant A) \leqslant P(V \geqslant A), \tag{15}$$

where V is a random variable with $E(V^2) < \infty$.

Roughly speaking, the arrival times of a weakly dependent input differ from the events of a renewal process by amounts whose distributions are all dominated by one with finite second moment. An immediate deduction from the preceding discussion is

Theorem 4. Let a sequence of queues Q_α, satisfying $\mathscr{A}$ and equation (11), have a weakly dependent input, and service times independent of the input process and of each other. Then, if, for some $b > 0$, $E(e^{bs}) < \infty$, Q_α has the heavy traffic property.

5. Examples of Queues with Weakly Dependent Inputs

5.1. *Queues in Tandem*

Suppose that, before arriving in the queue, the customers have to pass through some other system, such as one or more queues, which imposes a delay on each customer, but does not change the order of arrival. Then, if the inter-arrival times at this system are τ_n, and if C_n suffers a delay d_n, we have

$$t_n = \tau_n + d_{n+1} - d_n. \tag{19}$$

Hence, if the inter-arrival times τ_n are independent, if the delaying system is stable in the sense that the sequence $\{d_n\}$ is stochastically dominated by a variable D, and if $E\tau$ and ED^2 are finite, then the input $\{t_n\}$ is weakly dependent. Thus, in heavy traffic, the limiting waiting-time distribution is the same as if the delaying system were absent. This means that if we have a number of queues in tandem, and if one of them is near its critical state, then we can examine its behaviour, to a first approximation, independently of the other queues.

5.2. *Scheduled Arrivals*

Suppose that the customers are due to arrive at unit intervals, but that each customer C is late by an amount $e(C)$, where the $e(C)$ are independent and identically distributed. We allow the customers to be so late that they may arrive in an order other than that in which they were scheduled to arrive. Loynes (1962b) has shown that, for this case,

$$t_n = 1 + g_n - g_{n-1}, \tag{20}$$

where the integral part of g_n is the number of customers overdue when C_{n+1} arrives.

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Hence, in order to show that we have a weakly dependent input, it suffices to show that the number $\nu(m)$ of customers overdue just after the mth scheduled arrival is dominated by a random variable with finite second moment. Now $\nu(m)$ is the number of inequalities

$$e_j > j \quad (j = 0, 1, \ldots, m-1)$$

which are satisfied, where e_j is the lateness of the customer due to arrive j units of time before the time we are considering. Hence the distribution of $\nu(m)$ is that of the number of successes in m independent trials with probabilities of success

$$p_j = P(e > j).$$

It is known that, if

$$\sum_1^\infty p_j < \infty,$$

the number of successes converges in distribution to a random variable ν which has all its moments finite. If $\Sigma p_j = \infty$ it means that $\nu(m) \to \infty$ as $m \to \infty$, and we do not obtain an arrival pattern tending to stationarity. It is clear that the condition $\Sigma p_j < \infty$ is equivalent to $E(e) < \infty$. Hence, if e has finite mean we have a weakly dependent input, and the lateness of customers has, in heavy traffic, vanishingly small effect. Note that, in this case, equation (7) becomes

$$E(w) \simeq \operatorname{var}(s)/\{1 - E(s)\},$$

and the approximation will be valid when $1 - E(s)$ is small compared with the standard deviation of s.

There are many other more complicated problems in which an input which is initially an independent one is modified in some way before entering the queue. Such problems can often be shown to have weakly dependent inputs, and hence to lead to the heavy traffic property.

6. The Queue Length and the Busy Period

The analysis so far has been in terms of the waiting time, but we can also examine other distributions in the same way. We here discuss briefly the argument and results for two cases of interest, the queue length and the busy period.

If we suppose that the service times are independent, then the waiting time of a customer consists of n service times, together with a part s' of a service time; n being the length of the queue on the arrival of the customer. If we have a sequence of queues with the heavy traffic property, then (taking, for convenience, $\sigma(\alpha) = 1$),

$$E(e^{-\alpha\theta w(\alpha)}) \to (1 + \tfrac{1}{2}\theta)^{-1}.$$

Hence, since $\alpha s' \to 0$ in probability,

$$\begin{aligned}(1 + \tfrac{1}{2}\theta)^{-1} &\sim E\{E(e^{-\alpha\theta s})\}^n \\ &\sim E\{E(1 - \alpha\theta s)\}^n \\ &\sim E(e^{-\alpha\theta\bar{s}n}) \quad \text{if} \quad \bar{s} = E(s).\end{aligned}$$

Thus, if $\bar{s}$ tends to a limit as $\alpha \to 0$, then αn is asymptotically exponentially distributed with mean equal to the limiting value of $E(w)/E(s)$.

In the case of the busy period, however, asymptotic results of this type are not possible. To see this, consider the simplest case of all, namely that in which the arrivals

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form a Poisson process and the service times have an exponential distribution. Under these assumptions it is known (Kendall, 1951) that the busy period Z has moment generating function

$$\Phi(\theta) = E(e^{-\theta Z}) = 2[1+\rho+\theta+\sqrt{\{(1+\rho+\theta)^2-4\rho\}}]^{-1} \tag{21}$$

in which we have taken $E(s) = 1$, $E(t) = \rho^{-1}$. As $\rho \to 1$ this tends to

$$[1+\tfrac{1}{2}\theta+\sqrt{\{(1+\tfrac{1}{2}\theta)^2-1\}}]^{-1}. \tag{22}$$

If $\theta = i\lambda$ (λ real) this is a continuous function of λ at $\lambda = 0$, and hence represents the moment generating function of a valid random variable. However, none of the moments of this variable are finite. In fact, it is not difficult to show that, for any integer n,

$$E(Z^n) \text{ tends to infinity with } (1-\rho)^{-2n+1}. \tag{23}$$

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